The Jones Polynomial: Quantum Algorithms & Applications in Quantum Complexity Theory

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Organization of the talk

- Short introduction to quantum computing
- Links as closures of braids
- Jones polynomial
- Jones-Wenzl representations of the braid group
- Representation-theoretic formula for the Jones polynomial
- Quantum computation as approximate evaluation of the Jones polynomial

Quantum computational power

in words:

- the quantum computer is very efficient at multiplying certain big sparse unitary matrices and estimating entries of the resulting product
- this is the hardest problem which can be efficiently solved by a quantum computer

everything is made more precise by the quantum circuit model

Quantum circuit model

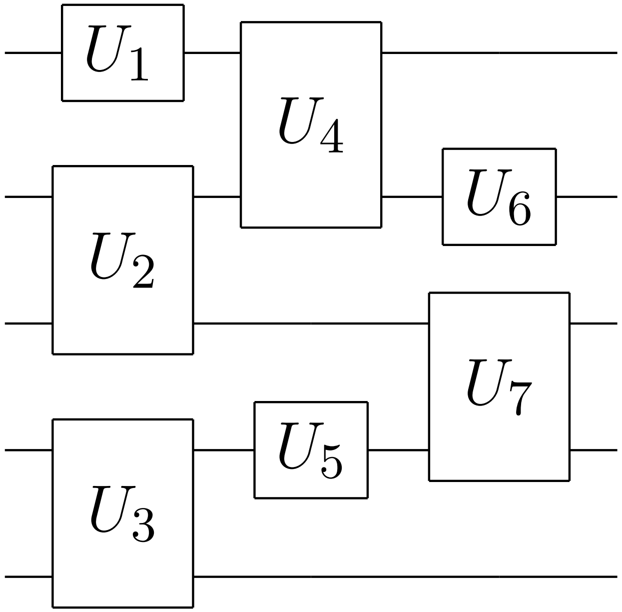
- let $\mathcal{H}$ be the Hilbert space given by the n-fold tensor product of $\mathbb{C}^2$

$$\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2$$

  and $SU(2^n)$ be the special unitary group acting on $\mathcal{H}$
- $SU(2^n)$ is generated by embeddings of unitary matrices acting on $\mathbb{C}^2$ (single qubits) and $\mathbb{C}^2 \otimes \mathbb{C}^2$ (two qubits); these are refered to as single qubit and two qubit *gates*
- the *complexity* $\kappa(U)$ of $U \in SU(2^n)$ is the minimal number of gates such that their product gives U
- the complexity $\kappa_\epsilon(U)$ is the minimal number of gates such their product yields V and

$$\|U - V\| \leq \epsilon$$

Quantum circuit model



$$(I_2 \otimes U_6 \otimes U_7 \otimes I_2) \cdot (U_4 \otimes I_2 \otimes U_5 \otimes I_2) \cdot (U_1 \otimes U_2 \otimes U_3)$$

Universal set & Solovay-Kitaev theorem

a universal set $\mathcal{G}$ is a finite subset of $SU(4)$ (closed under taking inverses) such that the subgroup $\langle \mathcal{G} \rangle$ generated by $\mathcal{G}$ is dense in $SU(4)$

Solovay-Kitaev Theorem:
let $\mathcal{G}$ be a universal set and $\delta > 0$

an arbitrary $U \in SU(4)$ can be approximated to within a distance δ by a sequence of $\mathrm{polylog}(1/\delta)$ gates from $\mathcal{G}$

moreover, such a sequence be "compiled" by a classical computer in $\mathrm{polylog}(1/\delta)$ time

$\Longrightarrow$ the definition of complexity does not depend "too much" on the particular choice of $\mathcal{G}$

Approximate Circuit Matrix Entry

- let U be a unitary matrix given by a quantum circuit on n qubits with $m = poly(n)$ gates and $\epsilon = 1/poly(n)$
- estimate the first diagonal entry of U with precision ϵ, that is, output u such that

$$|u - \langle 00 \ldots 0|U|00 \cdots 0\rangle| \leq \epsilon$$

there is a quantum algorithm with running time which is only polynomial in n

the best classical algorithms have exponential running time

ACME is a complete problem for the quantum complexity class Bounded error Quantum Polynomial time (BQP)

Quantum circuit model

the quantum circuit model is a mathematical specification of what (we hope) quantum computer could do

it is justified by the fundamental laws of quantum mechanics

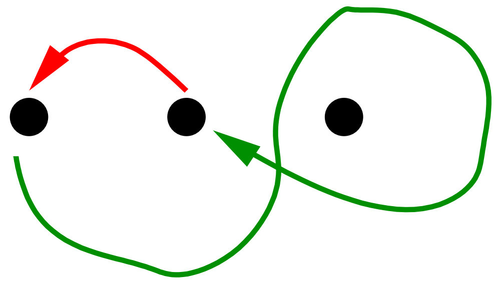
there are many different proposals for quantum hardware which implement this model

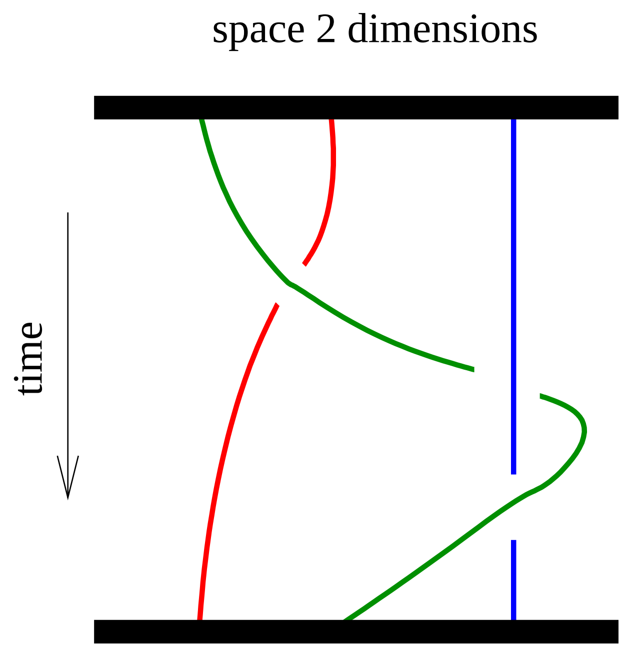
- ion traps
- optical lattices
- quantum dots
- nuclear magnetic resonance
- ...

a different model of computation is given by so-called topological quantum computation

Topological Quantum Computing

particle exchanges in a 2+1 dimensional quantum system of n identical particles are governed by the braid group B_n





particles move along the arrows, so that the original particle positions are once again occupied

the braid describes their trajectories in space-time

What happens when we braid particles?

depends on the type of particles

- bosons $\rightarrow$ nothing
- fermions $\rightarrow$ wavefunction acquires ± 1 factor
- abelian anyons $\rightarrow$ phase factor $e^{2\pi i \varphi}$
- non-abelian anyons $\rightarrow$ nontrivial transformation

  (quantum Hall quasi-particles and quasi-holes)

certain anyons could be used for realizing universal quantum computation with built-in fault-tolerance

fault-tolerance due to topological properties

Mathematical description

state space: some Hilbert space V

elementary operations: generators $\sigma_i^{\pm 1} \in B_n$

let $|\psi\rangle \in V$ be the state of the topological quantum computer

how does $|\psi\rangle$ change if we apply the elementary operation $\sigma_i^{\pm 1}$?

$$|\psi\rangle \mapsto \pi(\sigma_i^{\pm 1})\,|\psi\rangle$$

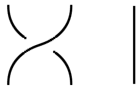
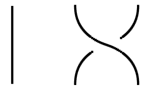
where π is a unitary representation of the braid group B_n

Braid group B_n

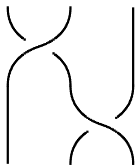
the braid group B_n on n strands is generated by $\sigma_1, \sigma_2, \ldots, \sigma_{n-1}$ satisfying

$$\begin{aligned} \sigma_i \sigma_j \sigma_i &= \sigma_j \sigma_i \sigma_j \quad && |i-j| = 1 \\ \sigma_i \sigma_j &= \sigma_j \sigma_i && |i-j| \geq 2 \end{aligned}$$

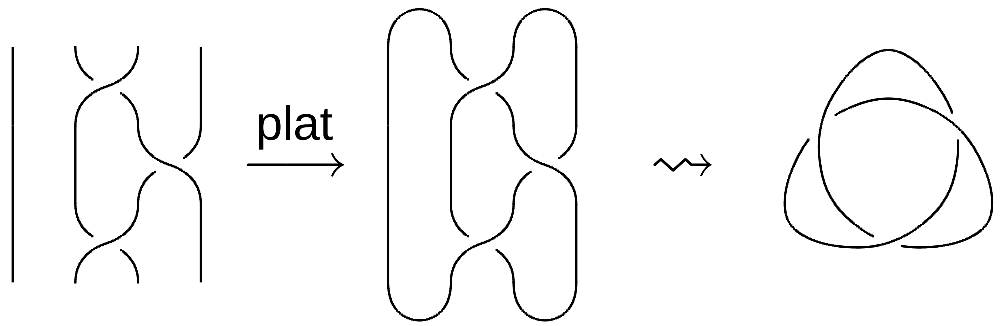
for instance, the braids $\sigma_1, \sigma_2^{-1} \in B_3$ are given by

$\sigma_1 =$  and $\sigma_2^{-1} =$ 

the group product then corresponds to vertical concatenation of braids, so that

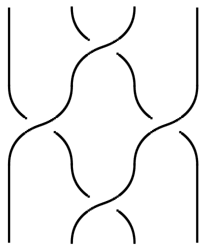
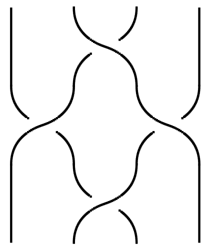
$\sigma_2^{-1} \sigma_1 =$ 

Links as plat closures of braids



Birman moves

two braids b and b' have isotopic closures iff we can go from b to b' by applying a sequence of Birman moves

$\sigma_{2i}\sigma_{2i+1}\sigma_{2i-1}\sigma_{2i} =$  , $\sigma_{2i}\sigma_{2i+1}\sigma_{2i-1}^{-1}\sigma_{2i}^{-1} =$  , $\sigma_{2i-1}^{\pm 1}$

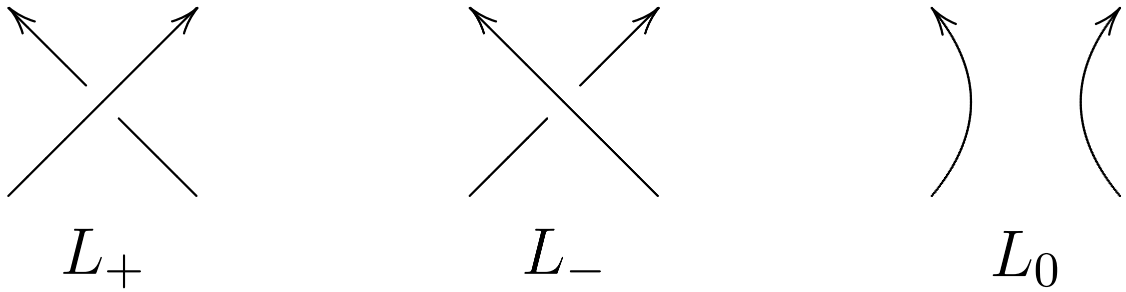
first Birman move: $b \mapsto ubv$, where u, v are elements of the Birman subgroup generated by the above braids

second Birman move: $b \mapsto \sigma_{2m}^{\pm 1} b \in B_{2m+2}$, where two strands are added to the right of b before multiplying with σ_{2m}

References

- V. F. R. Jones, “Hecke algebra representations of braid groups and link polynomials,” *Annals of Mathematics*, vol. 126, pp. 335-388, 1987.
- H. Wenzl, “Hecke algebras of type A_n and factors,” *Inventiones Mathematicae*, vol. 92, pp. 349-383, 1988.

Jones polynomial



denote three oriented links that differ in a small region, as symbolized by the above diagrams, but are otherwise the same

the Jones polynomial $J_L(t)$ is a invariant of oriented links that satisfies the skein relation

$$t^{-1}J_{L_+}(t) - tJ_{L_-}(t) = (t^{1/2} - t^{-1/2})J_{L_0}(t)$$

Computation of the Jones polynomial

with the help of the Kauffman bracket

$$\left\langle \times \right\rangle = A \left\langle)\,(\right\rangle + A^{-1} \left\langle \asymp \right\rangle$$

the Jones polynomial is

$$J_L(t) = A^{3w(\overrightarrow{L})} \langle L \rangle \Big|_{A=t^{1/4}}$$

where $w(L)$ is the writhe of L

connection between quantum computing and the Jones polynomial is based on

the Jones polynomial of plat closures evaluated at roots of unity can be expressed with the help of a certain irrep of B_n

Hecke algebra $H_n(q)$ - deformation of $\mathbb{C}S_n$

Hecke algebra $H_n(q)$ is generated by $g_1, \ldots, g_{n-1}$ satisfying the relations

$$\begin{array}{rcll} g_i^2 & = & g_i(q-1)+q & \\ g_i g_j g_i & = & g_j g_i g_j & |i-j|=1 \\ g_i g_j & = & g_j g_i & |i-j|>1 \end{array}$$

this is a deformation of the symmetric group, which is obtained when $q = 1$

we are interested in the case $q = e^{2\pi i/l}$, q is lth primitive root of unity

the map $B_n \to H_n(q)$ defined by $\sigma_i \mapsto g_i$ is a representation of the braid group

Quantum integers

define the quantum integers

$$[d]_\ell = \left. \frac{q^{d/2} - q^{-d/2}}{q^{1/2} - q^{-1/2}} \right|_{q=e^{2\pi i/\ell}} = \frac{\sin(\pi d/\ell)}{\sin(\pi/\ell)}$$

the loop value is

$$D \equiv [2]_\ell = q^{1/2} + q^{-1/2} \Big|_{q=e^{2\pi i/\ell}} = 2\cos(\pi/\ell)$$

$H_n(q)$ in terms of projectors

define projectors $e_i = (q - g_i)/(1+q)$ by setting

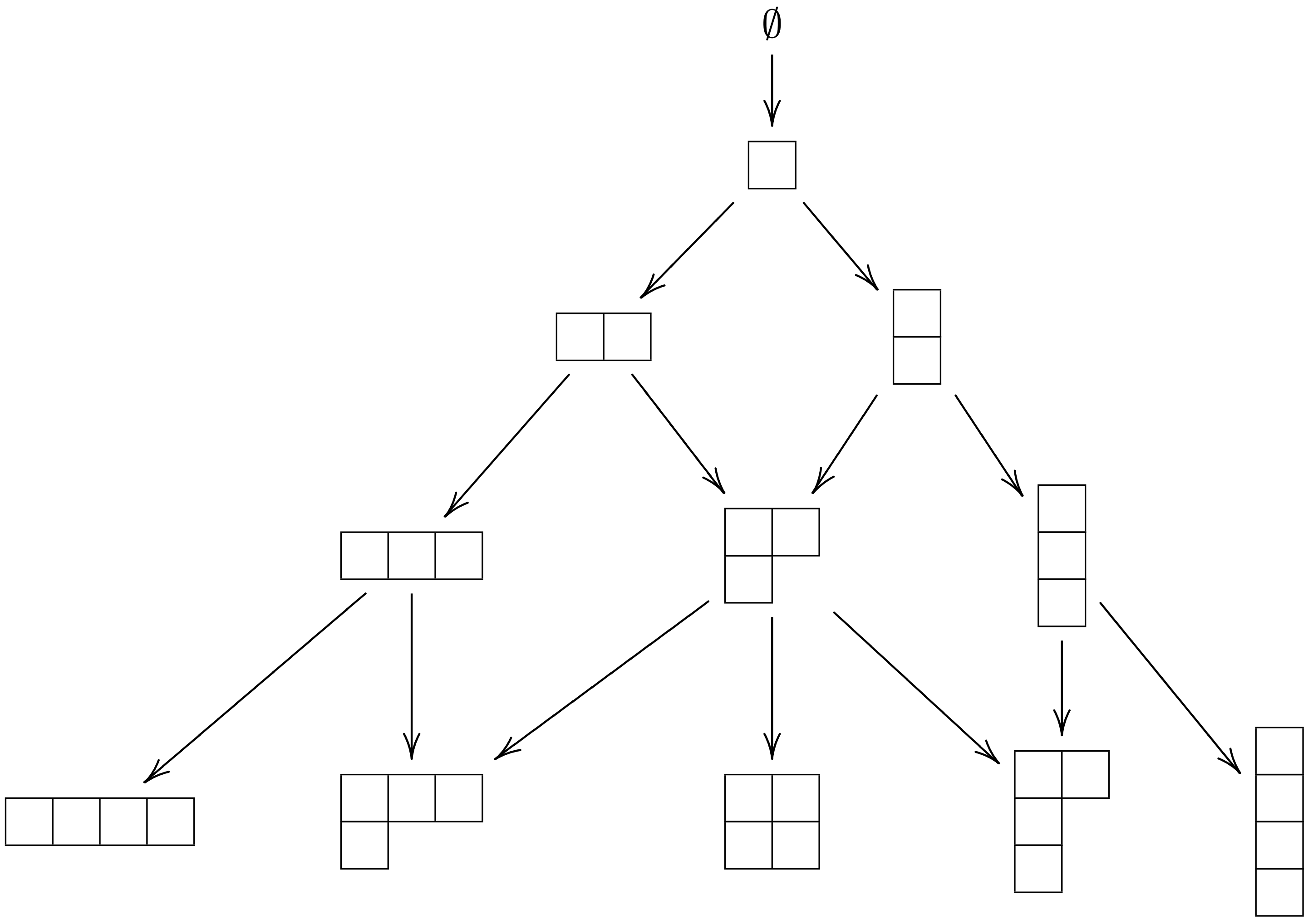
$$g_i = q(1 - e_i) - e_i = q - (1+q)e_i$$

using these projectors, the relations of $H_n(q)$ may be expressed as

$$\begin{array}{rcll} e_i^2 & = & e_i & \\ e_i e_j e_i - \tau e_j & = & e_j e_i e_j - \tau e_i & \qquad |i-j| = 1 \\ e_i e_j & = & e_j e_i & \qquad |i-j| > 1 \end{array}$$

where $\tau = D^{-2}$

Young lattice Λ_n



Standard tableaux T_n

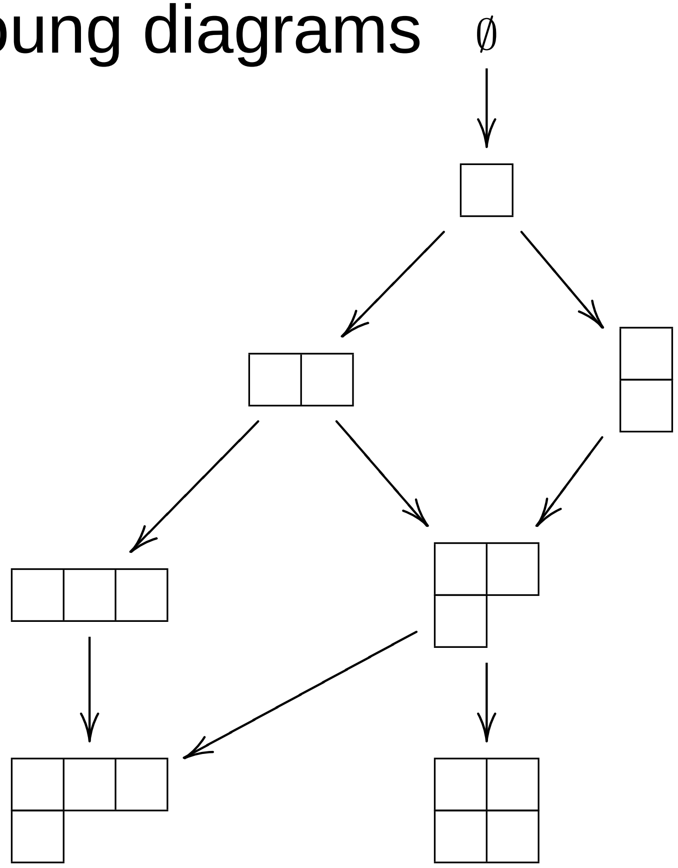
a standard tableau T_λ is an assignment of the numbers $1, 2, \ldots, n$ to the boxes of $\lambda \in \Lambda_n$ such that the numbers increase in each row and each column of λ

standard tableaux correspond to paths on the Young lattice

truncated Young lattice $\Lambda_n^{(k,l)}$

of (k, l)-admissible Young diagrams

for example, $\Lambda_4^{(2,5)}$



$$\Lambda_n^{(k,\ell)} = \{\lambda \in \Lambda_n : \lambda_{k+1} = 0, \lambda_1 - \lambda_k \leq \ell - k\}$$

(k,ℓ)-admissible standard tableaux $T_n^{(k,\ell)}$

correspond to paths on $\Lambda_n^{(k,\ell)}$

Jones-Wenzl irreps of $H_n(q)$ and B_n

enumerated by (k, ℓ)-admissible Young diagrams $\lambda \in \Lambda_n^{(k,\ell)}$

corresponding irrep $\pi_\lambda^{(k,l)}$ acts on the vector space

$$V_\lambda^{(k,\ell)} = \mathrm{span}\{|t\rangle \,:\, t \in T_n^{(k,\ell)}\}$$

whose basis vectors are the (k, ℓ)-admissible standard tableaux of shape λ

define $s_i(t)$ to be the standard tableau obtained form t by exchanging the positions of i and $i + 1$ (not valid, set $s_i(t) = t$)

define hook length $d_i(t) = c_i(t) - c_{i+1}(t) - (r_i(t) - r_{i+1}(t))$

Jones-Wenzl irreps continued

let $V_{t,i}^{(k,l)}$ be the subspace of $V_\lambda^{(k,\ell)}$ spanned by $|t\rangle$ and $|s_i(t)\rangle$

define $\quad a_\ell(d) = \frac{[d+1]_\ell}{[2]_\ell [d]_\ell}$

the image $\pi_\lambda^{(k,\ell)}(e_i)$ of e_i under $\pi_\lambda^{(k,\ell)}$ acts as

$$\pi_\lambda^{(k,\ell)}(e_i) = \begin{pmatrix} \frac{a_\ell(d_i(t))}{\sqrt{a_\ell(d_i(t))(1-a_\ell(d_i(t)))}} & \sqrt{a_\ell(d_i(t))(1-a_\ell(d_i(t)))} \\ & 1 - a_\ell(d_(i(t)) \end{pmatrix}$$

on $V_{t,i}^{(k,\ell)}$ and as identity on the orthogonal complement

Jones-Wenzl irreps continued

corresponding irrep of the braid group B_n is defined by

$$\pi_\lambda^{(k,\ell)}(g_i) = q\,\mathbf{1}_{V_\lambda^{(k,l)}} - (1+q)\pi_\lambda^{(k,\ell)}(e_i)$$

Representation-theoretic formula

let $\lambda = [m, m]$ be the “rectangular” Young diagram with two rows, each having m boxes

define the normalized vector

$$|\varphi_{2m}\rangle = \left| \begin{array}{|c|c|c|}\hline 1 & 3 & 5 \\ \hline 2 & 4 & 6 \\ \hline \end{array} \cdots \begin{array}{|c|}\hline 2m{-}1 \\ \hline 2m \\ \hline \end{array} \right\rangle$$

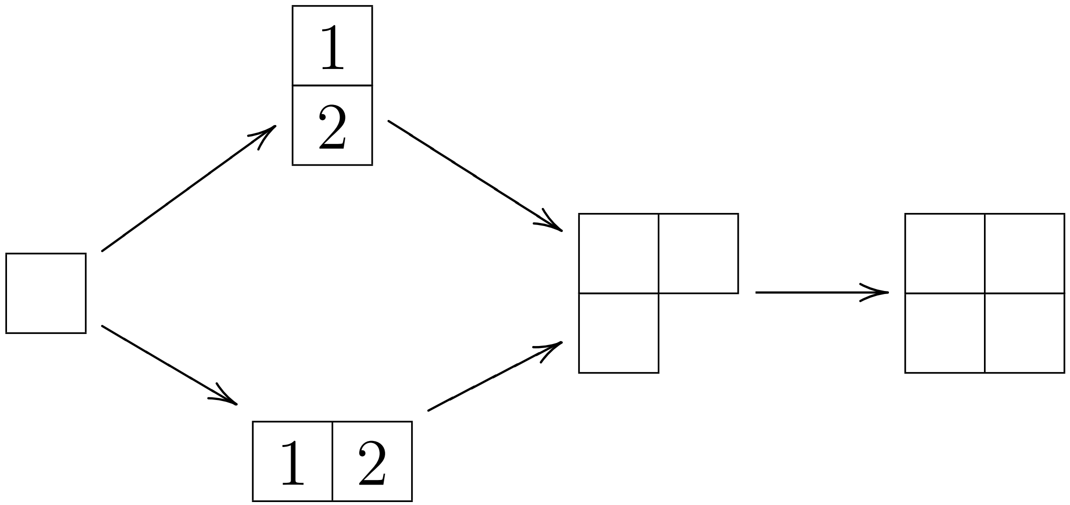
Representation-theoretic formula

the norm of the Jones polynomial of the plat closure of any $b \in B_{2m}$ evaluated at $e^{2\pi i/l}$ is given by

$$D^{m-1} \left| \langle \varphi_{2m} | \, \pi_{\lambda}^{(2,l)}(b) \, | \varphi_{2m} \rangle \right|$$

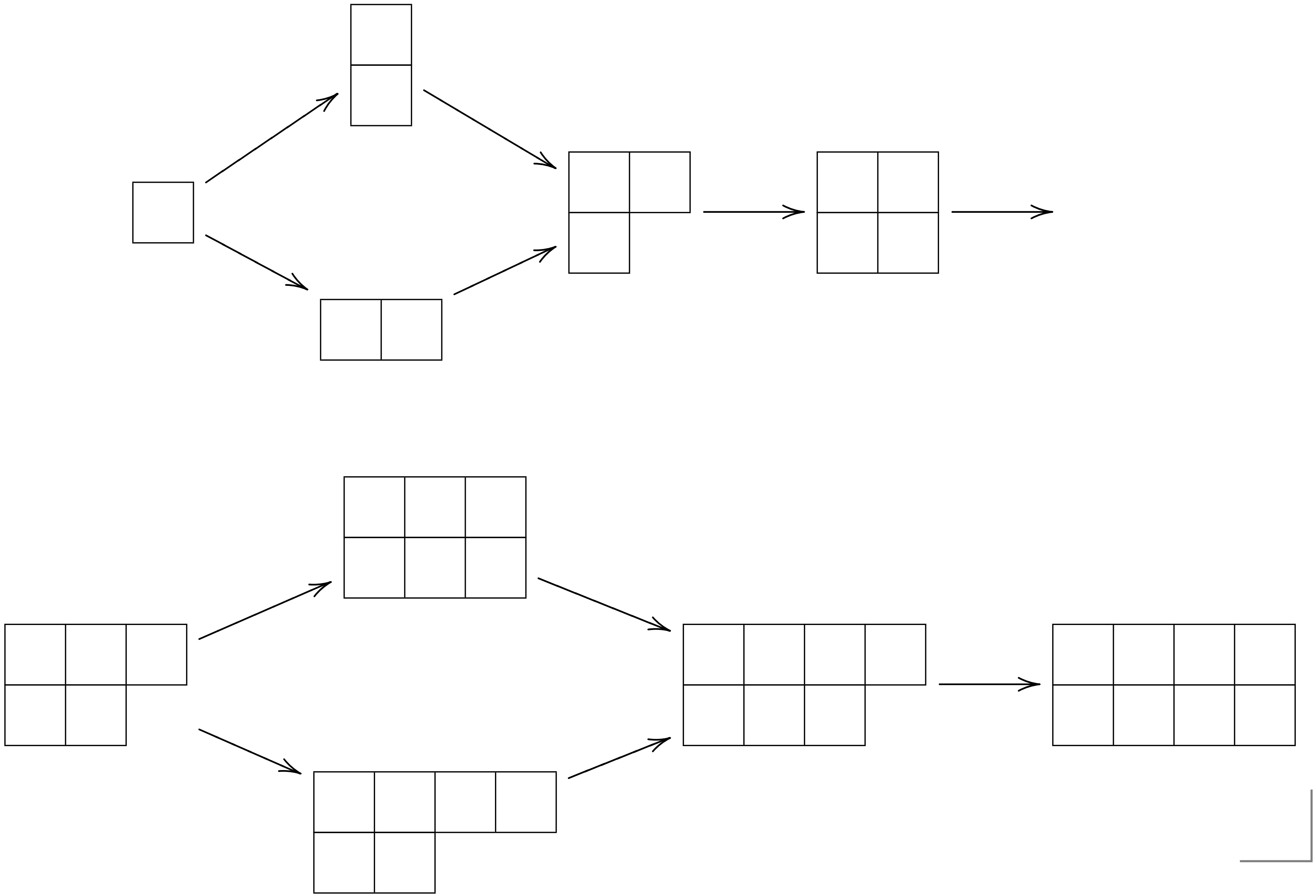
has the same structure as $| \langle 00 \ldots 0 | \, U \, | 00 \ldots 0 \rangle \, |$ in the BQP-complete problem “Approximate circuit matrix entry”

Our encoding of a qubit in 4 strands



$$|0\rangle \equiv \left| \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & 4 \\ \hline \end{array} \right\rangle \qquad |1\rangle \equiv \left| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & 4 \\ \hline \end{array} \right\rangle$$

Our encoding of 2 qubits in 8 strands



Encoding of 2 qubits into 8 strands

four periodic structure

$$|t_0\rangle\,|t_0\rangle \quad\mapsto\quad \left|\,\begin{array}{|c|c|c|c|}\hline 1&3&5&7\\\hline 2&4&6&8\\\hline\end{array}\right\rangle$$

$$|t_0\rangle\,|t_1\rangle \quad\mapsto\quad \left|\,\begin{array}{|c|c|c|c|}\hline 1&3&5&6\\\hline 2&4&7&8\\\hline\end{array}\right\rangle$$

$$|t_1\rangle\,|t_0\rangle \quad\mapsto\quad \left|\,\begin{array}{|c|c|c|c|}\hline 1&2&5&7\\\hline 3&4&6&8\\\hline\end{array}\right\rangle$$

$$|t_1\rangle\,|t_1\rangle \quad\mapsto\quad \left|\,\begin{array}{|c|c|c|c|}\hline 1&2&5&6\\\hline 3&4&7&8\\\hline\end{array}\right\rangle$$

$\mathcal{C}^{(2)} =$ span$\{|t_0\rangle\,|t_0\rangle, \ldots, |t_1\rangle\,|t_1\rangle\}$ is a subspace of $V_{\yng(4,4)}$

$\mathcal{C}^{(n)} =$ span$\{|t_0\rangle, |t_1\rangle\}^{\otimes n} \leq V_{\yng(2,2)\cdots\yng(1,1)}$

Our encoding of two-qubit gates

makes use of

Theorem [Freedman, Larsen, Wang]

the image of the eight strand braid group B_8 under the irreducible representation $\pi_{\yng(4,4)}$ is dense in the special unitary group $SU\left(V_{\yng(4,4)}\right)$

$$\Downarrow$$

any two-qubit gate U can be efficiently approximated by sequences elementary operations $\sigma_1^{\pm 1}, \ldots, \sigma_7^{\pm 1}$

such sequences can be efficiently found by a classical computer (Solovay-Kitaev theorem)

Efficient encoding of quantum circuits

Theorem [W. and Yard]

let $U = U_m U_{m-1} \cdots U_1$ be a quantum circuit of length m acting on n qubits and let $\epsilon > 0$

then there is a braid $b \in B_{4n}$ with $poly(m, log(1/\epsilon))$ crossings such that

$$\| U - \pi(b)|_{\mathcal{C}^{(n)}} \|_\infty \leq \epsilon$$

$\pi(b)|_{\mathcal{C}^{(n)}}$ denotes restriction to computational subspace

moreover, such a braid can be found from a description of the circuit in $poly(m, log(1/\epsilon))$ time on a classical computer

Putting everything together

- let $\mathcal{G}$ be any universal set and $\epsilon = 1/poly(n)$
- let U be a quantum circuit acting on n qubits and using $m = poly(n)$ gates from $\mathcal{G}$

given U and ϵ we can efficiently construct a braid $b \in B_{4n}$ of length $poly(n)$ such that

$$\Big| \, | \langle 00 \ldots 0 | \, U \, | 00 \ldots 0 \rangle \, | - | \, \langle \varphi | \, \pi^{(2,l)}(b) \, | \varphi \rangle \, | \, \Big| < \epsilon/2$$

$$\Big| \, | \langle 00 \ldots 0 | \, U \, | 00 \ldots 0 \rangle \, | - D^{-2n+1} | J_{\hat{b}}(e^{2\pi i/l}) | \, \Big| < \epsilon/2$$

where $J_{\hat{b}}(e^{2\pi i/l})$ is the Jones polynomial of the plat closure of b evaluated at the root of unity $t = e^{2\pi i/l}$

Putting every thing together continued

if we can estimate $|J_{\hat{b}}(e^{2\pi i/l})|$ with precision $\epsilon \cdot D^{2n-1}/2$, then we can solve the BQP-complete problem “approximate circuit matrix entry”

this shows that approximately evaluating the Jones polynomial at roots of unity is BQP-hard

More results

- Approximately evaluating the Jones polynomial of the plat and trace closure of braids are in BQP
- Approximately evaluating the HOMFLYPY polynomial of trace closure of braids at certain pairs of values is in BQP
- Simplified and new proofs of the computational complexity of certain problems concerning the Jones polynomial

Historical overview

- “Approximately evaluating the Jones polynomial is BQP-complete” proved in the context of topological quantum computation; uses topology and quantum field theories [Freedman, Kitaev, Larsen, Wang]
- “Approximately evaluating the Jones polynomial is in BQP”; simplified proof within the quantum circuit model [Aharonov, Jones, Zhang]
- “Approximately evaluating the Jones polynomial is in BQP-hard”; simplified proof based on a certain encoding of quantum circuits in the rectangular representation of the braid group [W. and Yard]